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A Study on Memory Stack Process by Hybrid Copper Bonding (HCB) Technology

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***Abstract*— Hybrid copper bonding (HCB) technology has been introduced and evaluated to overcome a fine pitch limit and degradation in thermal properties in 3D semiconductor package structure. In this study, we successfully demonstrated 3D memory stacking package by HCB technology which integrated Cu-Cu diffusion bonding along with oxide bonding simultaneously. HCB process was developed and die-to-wafer (D2W) multi-stack structure was realized which shows excellent electrical and thermal properties. The flatness of the bonding surface was controlled by chemical mechanical polishing (CMP), and highly activated bonding surface desirable for HCB was obtained through optimization of plasma condition. The properties of the chip surface were measured in terms of oxide topography, Cu dishing height, surface hydrophilicity, hydroxyl group density, and adhesion force. Then package chips were stacked up to 12 levels by HCB method to demonstrate 12 stacked D2W package. In particular, we focused on eliminating oxide voids at the bonding interface since even a single void can deteriorate the functional properties of the whole stacked package. Oxide voids were classified into several categories on the basis of the causes, then the sources of the void were eliminated in advance to achieve void free HCB interfaces. Through the accomplishment of D2W HCB technology, it is expected to be applied to advanced 2.5D/3D packages for superior thermal and electrical performances.**

Keywords- Hybrid bonding, Die to Wafer, Multi-stack, TSV, Electrical and thermal performance

I. INTRODUCTION

Recently, advanced package technology has been rapidly emerged as one of the technologies that can overcome the limits of Moore's Law and improve device performance dramatically. It requires increased number of input/output interconnection density within reduced package size resulting in size reduction in bump and pitch. However, conventional micro solder bumps have fine pitch limitation due to bump bridging issues by melted solders with narrower bump pitch. They also have degraded thermal conductivities by polymer materials existing in the bump gaps. It gets more severe in high integration density applications such as 3D memory package or heterogeneous 2.5D packages which demand high power delivery and thermal dissipation.

To improve the technical difficulties mentioned above, a HCB technology has been introduced [1-6]. Unlike the conventional flip chip or thermo-compression bonding, the HCB does not require solder bumps and gap filling epoxy materials. For HCB, chemical vapor deposition (CVD) oxide layers replace micro solder bumps and polymer adhesives, enabling a gapless D2W structure [7, 8]. For HCB, hydrophilic surface with hydroxyl group is obtained by plasma activation and hydration process. The Si oxide layers on which the hydroxyl group is formed are bonded by Van der Waals force and hydrogen bonding, thereafter transiting to oxygen covalent bond by the annealing process [9, 10]. Strong adhesion is maintained by oxide bonding and electrical interconnection is made via Cu-Cu direct bonding. Absence of polymer in the gapless D2W structure improves thermal properties by reducing thermal resistance at the bonding interface. In addition, excellent electrical properties are expected because shortest electrical conduction paths could be achieved through Cu-Cu direct bonding. In fact, the HCB is not a new technology. It has been studied in wafer level package where wafer-on-wafer (WOW) bonding method has been used [11, 12]. However, it is very limited to apply WOW bonding method to 3D stacking structure since only known good dies should be built vertically. Therefore, the D2W HCB technology has been emerged as a promising technology for 3D chip stacking package.

In this study, we demonstrated a 12 stacked D2W package for 3D package through establishing key technologies such as planarization and activation of bonding surface, and multi-chip stacking process technologies. Optimal properties of bonding surface could be obtained by modifying surface topography, Cu pad dishing height variation and reactive functional group on the bonding surface for test vehicles. Thereby, we have achieved void free bonding interfaces, excellent electrical conductivity and low thermal resistance for vertically multi-chip packages.

II. STRUCTURE AND PROCESS

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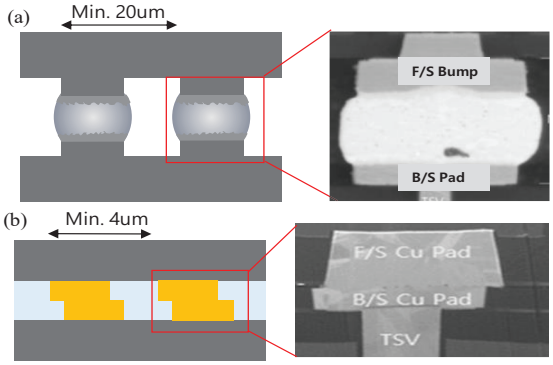

Fig. 1 Comparative joints between (a) micro solder bumps using conventional flip chip bonding and (b) Cu to Cu pads using HCB.

Figure 1 shows the comparative joints between conventional flip chip and Cu to Cu direct bonding structures. As shown in figure 1(a), the traditional flip chip bonding structure comprises solder bumps and epoxy materials at the joint gap. However, the HCB technology can realize a gapless D2W structure shown in figure 1(b) where micro solder bumps and gap filling materials are unnecessary. Instead, HCB requires extremely flat and clean chip surface for defect-free bonding interface. A typical process flow for HCB D2W can be seen in figure 2. Oxide layers were introduced on the surfaces of top and base wafers by CVD, and then the oxide layers were planarized by means of optimized CMP process [13], thereby obtaining desirable oxide surface topography and Cu pad dishing height. Next, the top wafer was diced into single chips after coating protective layer on the wafer surface to prevent any particle and contaminant that cause interface voids during the chip bonding process. To activate the bonding surface, plasma and hydration processes were conducted with the process condition for better hydrophilicity and higher density of hydroxyl group on the bonding surface, followed by stacking chips vertically up to 12-hi via HCB method and placing the wafer with stacked chips into an high temperature annealing chamber for covalent bonding between oxide layers and metallic bonding by Cu-Cu contact and diffusion of Cu atoms [14-16].

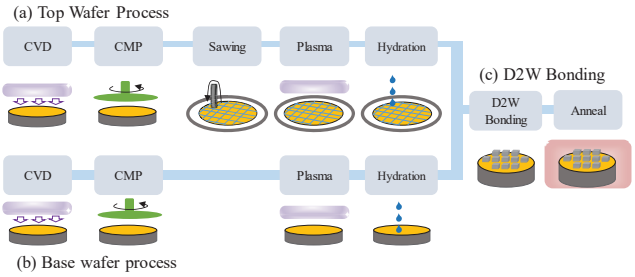

Fig. 2 D2W HCB Process. (a) Top wafer for stacked chips, (b) Base Wafer, and (c) D2W Stacking process.

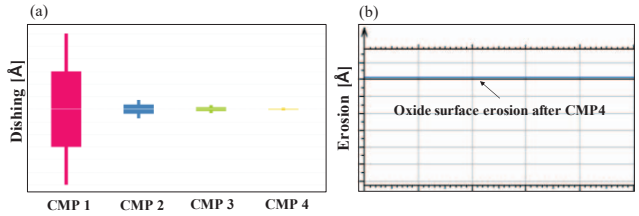

Fig. 3 CMP condition effects on (a) Cu pad dishing height variation, (b) oxide surface topography by applying CMP 4 process

III. RESULTS

A. Preparation of the Bonding Surface for HCB

Figure 3 shows the results of Cu dishing and oxide erosion according to CMP process condition. Modifying Cu pad and oxide surface topographies is the key technology of voids at the bonding interface [17]. Also, each Cu pad requires ultra-narrow height variation to accommodate Cu-Cu contact during a high temperature annealing step for good electrical connectivity. As shown in figure 3(a) and (b), Cu dishing height variation was reduced dramatically with applying CMP condition 4, which was attributed to the void-free bonding interface that is explained in section C. Figure 4 shows the graphs of correlation between contact angle and hydroxyl group density according to three conditions of plasma process. The lowest contact angle and the highest hydroxyl functional group density were acquired with plasma 1 condition as shown in figure 4(a), which was attributed to the highest oxide adhesion force as shown in figure 4(b). Furthermore, as shown in figure 5, plasma condition 1 showed the lowest Cu oxide thickness by controlling plasma condition and reducing the reaction rate between Cu and O, which could ensure Cu to Cu diffusion bonding for good electrical connectivity of D2W HCB package.

B. Optimization of HCB process for void-free oxide interface

It can be seen in figure 6 that HCB voids were generated in various types by diverse factors such as equipment and tool parts, package processing environment, or process conditions. Different types of void defects were observed by SAT analysis depending on the root causes. For example, particles with a size of 1um or bigger that generated during assembly

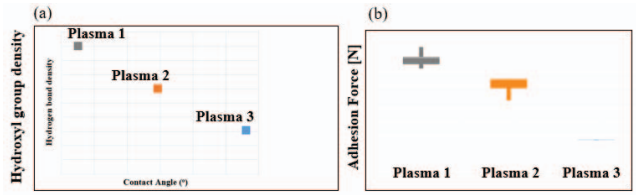

Fig 4. Plasma effects on (a) contact angle, hydroxyl group density and (b) adhesion force between oxide surfaces

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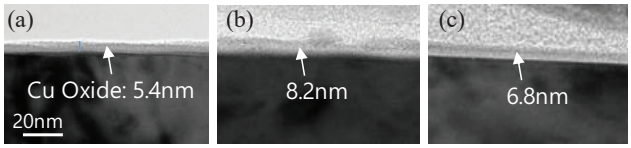

Fig. 5. A beneficial effect of plasma condition on the Cu oxide thickness. Plasma 1 showed the lowest Cu oxide thickness of 5.4nm (a), while the Cu oxide thickness was increased to 8.2 and 6.8nm at plasma 2 and plasma 3 as shown in (b) and (c).

process are fatal enough to form voids for 100%. Voids could also remain by trapped air or outgassing from various sources in the package structure, and it is almost impossible to remove voids once they are formed. It is essential to prevent void formation since even a single void can cause electrical failure and delamination of the interface during the subsequent package process and customer's usage environment. Figure 7 shows scanning acoustic tomography (SAT) images of the oxide interface conducted by each process condition 1-6. In order to prevent void formation, we reduced particles and contaminants dramatically by additional process such as protective layer coating (PLC), die transfer, PLC strip and cleaning and vision inspection steps before chip stacking process. Next, HCB conditions were optimized in terms of equipment and tool parts, and process parameters so as not to introduce any interfacial void during the chip bonding process. At condition 1, oxide voids were observed along the edge of the chip due to surface distortion along the edge line. At condition 2, a huge oxide void with more than 50% of the chip size area was observed at the center of the chip because of inappropriate process parameters, and the size of the void reduced by controlling the process parameters (condition 3). With further adjustment of parameters, the oxide void was divided into smaller oxide voids and located at corners of the chip (condition 4) or in the middle of the chip along x-axis condition 5). Finally, we could suppress the formation of oxide void through optimization of HCB process (condition 6). Before demonstrating D2W HCB multi-chip structure, we established key properties such as planarization of oxide surface and Cu dishing height by CMP, activation of bonding surface by modifying plasma and hydration process as mentioned above.

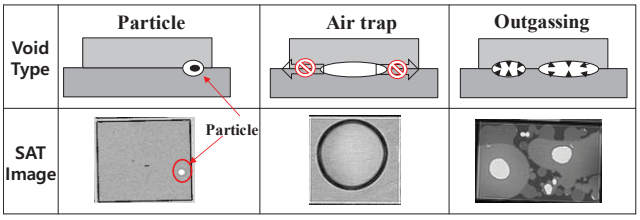

Fig. 6 Various types of oxide voids and their SAT images based on different causes: particle, air trap, and outgassing.

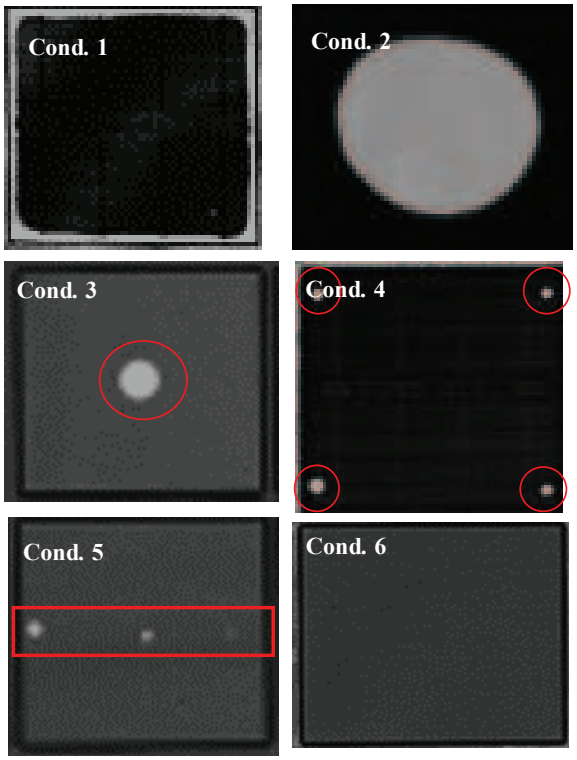

Fig. 7 SAT images at D2W bonding interface by different HCB conditions 1-6.

C. Demonstration of 12 stacked chips by HCB technology

Figure 8 shows void free SAT images of 12 stacked chip samples by incorporating optimized CMP, plasma and hydration process so as to have extremely flat chip surface and high density of hydroxyl group before multi-chip stacking process. As shown in figure 9, we could successfully demonstrate 12 stacked package structure by applying all the optimized process steps for D2W HCB. All 12 stacked chips were well bonded without any alignment failure, and each stacked chips were electrically connected through Cu pad and TSVs. Figure 10(a) shows a magnified cross-sectional images of a Cu to Cu joint after D2W bonding. Metallic bonding occurred through Cu expansion, Cu-Cu contact and Cu atom diffusion during a high temperature annealing step as shown in the lattice structure of Fig. 10(b). Although some discontinuous interface was found in the Cu to Cu joints, the electrical connectivity was confirmed that no circuit open or leakage failure occurred in 12 stacked chip packages.

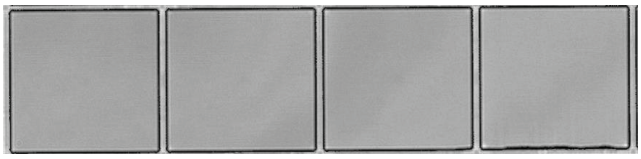

Fig. 8 Void free SAT images of 12 stacked D2W packages assembled by the optimized HCB condition 6.

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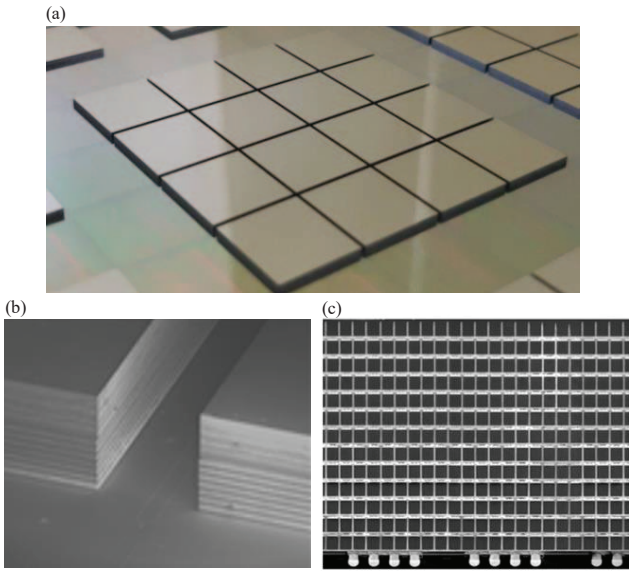

Fig. 9 Images of (a) a low magnification image, (b) a tilted SEM image, and (c) a cross-sectional SEM image of 12 stacked D2W.

D. Thermal Characterization of D2W HCB structure

Figure 11 shows that thermal resistance of HCB structure was 16.3% lower than that of conventional flip chip structure. Unlike the conventional micro bump structure, bumps and polymers between joint gaps are not required for the gapless D2W HCB structure. Accordingly, it was possible to maximize heat dissipation by reducing thermal resistance and lowering the operating temperature. It could prevent interfacial delamination caused by CTE mismatch between polymer and silicon as well.

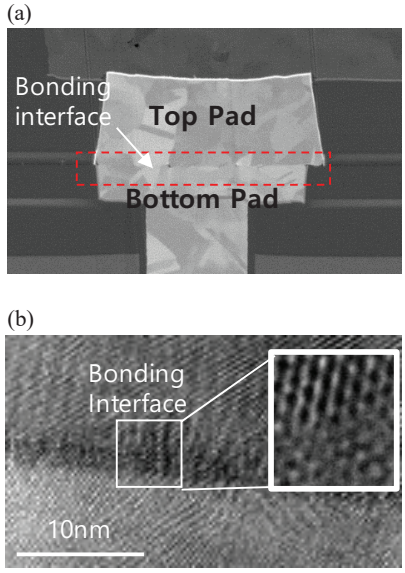

Fig. 10 Cross-sectional images of Cu to Cu bonding interface analyzed by (a) FIB and (b) HR-TEM.

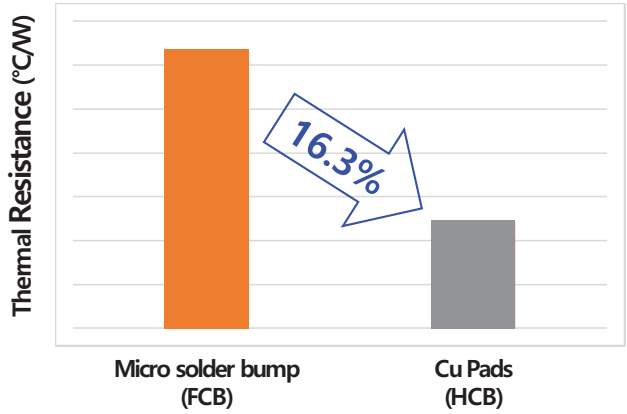

Fig. 11 Comparison of thermal resistance between micro solder bumps and Cu pads after fabricating 12 stacked chip package.

For further study in terms of thermal performance, we have been investigating the effect of oxide materials, joint gap structure, the number of stacks and chip design factors on the thermal resistance and working temperature of D2W HCB packages, which will point us in the right direction for the new bonding technology.

IV. CONCLUSION

In this paper, we demonstrated multi-stack D2W packages and reviewed feasibility results for the HCB technology. CMP process was optimized to control Cu dishing height variation and topography of oxide surfaces. Desirable Cu pad dishing height and oxide surface topographies were obtained for oxide-to-oxide bonding and Cu-Cu diffusion bonding. The plasma process condition was also optimized to modify the bonding surface for better hydrophilicity and higher hydroxyl functional group density. The optimized plasma condition also showed the lowest contact angle with the highest hydroxyl functional group density resulting in the highest oxide adhesion force. At the same time, it showed the lowest Cu oxide thickness which could ensure Cu to Cu diffusion bonding for good electrical connectivity. Finally, multi-stack HCB process was optimized to acquire a void free multi-chip stacking D2W structure. Void forming elements were eliminated and defect free 12-hi stacked D2W structures were successfully demonstrated by HCB technology.

Although we have achieved a D2W HCB structure by establishing overall processes including CVD, CMP, plasma and chip stacking process in this study, a lot of more challenging technologies should be developed for the mass production of advanced packages using HCB technology. In particular, we have yet to evaluate the electrical characteristics and reliability with active test chips. Therefore, we will study the effect of circuit configuration,

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bonding layer pattern density and underlying layers on the bonding interface quality, device performance and reliability with functional package devices.

References

[1] P. Ramm et el., "3D system integration technologies", Mat. Res. Soc. Symp. Proc. vol. 766, 2006.

[2] A. Phommahaxay et el., "Enabling Ultra-Thin Die to Wafer Hybrid Bonding for Future Heterogeneous Integrated Systems", 2019 IEEE 69th Electronic Components and Technology Conference (ECTC), 2019.

[3] B. Brandstätter et el., "High-speed ultra-accurate direct C2W bonding, 2020 IEEE 70th Electronic Components and Technology Conference (ECTC), 2020.

[4] G. Gao et el., "Scaling Package Interconnects Below 20µm Pitch with Hybrid Bonding, 2018 IEEE 68th Electronic Components and Technology Conference, 2018.

[5] T. Workman et el., "Die to Wafer Hybrid Bonding and Fine Pitch Considerations", 2021 IEEE 71st Electronic Components and Technology Conference (ECTC), 20201.

[6] F. Fournel et el., "Die to wafer direct bonding: from fundamental mechanisms to optoelectronic and 3D applications", 6th International Workshop on Low Temperature Bonding for 3D Integration (LTB-3D), DOI: 10.23919/LTB-3D.2019.8735248, 2019.

[7] C. Ko, K.N. Chen, "Wafer-level bonding/stacking technology for 3D integration", Microelectronics Reliability, Volume 50, Issue 4, Pages 481-488, 2010.

[8] G. Gao et el., "Development of Low Temperature Direct Bond Interconnect Technology for Die-To-Wafer and Die-To-Die Applications-Stacking, Yield Improvement, Reliability Assessment", 2018 International Wafer Level Packaging Conference (IWLPC), 2018.

[9] T. Fukushuma et el., "Oxide-Oxide Thermocompression Direct Bonding Technologies with Capillary Self-Assembly for Multichip-to-Wafer Heterogeneous 3D System Integration", Micromachines (Basel), Jul 2;10(7):445. doi: 10.3390/mi10070445, 2016.

[10] F. Inoue et el., "Influence of Composition of SiCN as Interfacial Layer on Plasma Activated Direct Bonding", ECS J. Solid State Sci. Technol. 8 P346, 2019.

[11] L. Peng et el., "Advances in SiCN-SiCN Bonding with High Accuracy Wafer-to-Wafer (W2W) Stacking Technology", 2018 IEEE International Interconnect Technology Conference (IITC), 2018.

[12] S. Kim et el., "Cu Microstructure of High Density Cu Hybrid Bonding Interconnection", 2019 IEEE 69th Electronic Components and Technology Conference (ECTC), 2019.

[13] T. Ueda et el., "A novel role for SiCN to suppress H2O outgas from TEOS oxide films in hybrid bonding", 2017 IEEE International Interconnect Technology Conference (IITC), 2017.

[14] J. Amandine et el., "Self-Assembly Process for 3D Die-to-Wafer using Direct Bonding: A Step Forward Toward Process Automatisation", 2019 IEEE 69th Electronic Components and Technology Conference (ECTC), DOI: 10.1109/ECTC.2019.00041, 2019.

[15] F. Inoue et el., "Protective Layer for Collective Die to Wafer Hybrid Bonding", 2019 International 3D Systems Integration Conference (3DIC), DOI: 10.1109/3DIC48104.2019.9058870, 2019.

[16] E. Bourjot et el., "Towards 5µm interconnection pitch with Die-toWafer direct hybrid bonding", 2021 IEEE 71st Electronic Components and Technology Conference (ECTC), 2021.

[17] E. Bourjot et el., "Towards a Complete Direct Hybrid Bonding D2W Integration Flow: Known-Good-Dies and Die Planarization Modules Development", 2019 International 3D Systems Integration Conference (3DIC), DOI: 10.1109/3DIC48104.2019.9058783, 2019.

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